```
---
title: "Impact of Hydroxyurea Treatment on the Progression and Outcomes of Sickle Cell Retinopathy: A Comparative Analysis"
subtitle: "The aims of this project are to assess whether hydroxyurea slows the progression of sickle cell retinopathy, reduces disease severity, and decreases the need for surgical interventions."
author: "Lomas Persad"
date: "`r format(Sys.Date(), '%B %d, %Y')`"

format:
  html:
    toc: true
    toc-depth: 3
    number-sections: False
    grid:
      margin-width: 150px
    self-contained: true
    theme: cosmo
    # css: styles.css
    # code-fold: true
    # code-tools: true
    # df-print: paged

fontsize: 14px

execute:
  echo: false
  warning: false

editor_options:
  chunk_output_type: console
---

## Data extraction

```{r}

library(tidyverse)
library(gtsummary)
library(flextable)
library(gt)
library(lme4)
library(lmerTest)
library(bit64)
library(emmeans)

library(survminer)
library(survival)

# load("data/report_data.RData")
load("data/report_data_subset.RData")

# names(eye_df)
# names(patient_df)
# names(va_timepoints)


# names(va_timepoints)




```


```{mermaid}

flowchart TD
    classDef title fill:#ffffff,stroke:#3a6ea5,stroke-width:2px,color:#000
    classDef step fill:#ffffff,stroke:#7f7f7f,stroke-width:1px,color:#000
    classDef exclude fill:#ffffff,stroke:#d9534f,stroke-width:1px,color:#000
    classDef final fill:#ffffff,stroke:#5cb85c,stroke-width:2px,font-weight:bold,color:#000
    classDef step stroke-width:1px,rx:6px,ry:6px;
    classDef exclude stroke-width:1px,rx:6px,ry:6px;
    classDef final stroke-width:1px,rx:6px,ry:6px;
%% ,font-weight:bold


    A[**SC retinopathy patients with available eye data in Cosmos**<br>5240 patients<br>04/02/2022 - 02/24/2026]:::title
    A --> B["<div style='width:340px; text-align:left'>
    <b>Exclusions</b><br>
    • 972 patients: Follow up < 6M <br>
    • 386 patients: Prior complication (H43.1x, H33.4x) <br>
    • 394 patients: missing demographic data <br>
    </div>"]:::exclude

       B--> C[**Final analytic cohort**<br>5775 eyes<br>3488 patients]:::final




    style C text-align:left



```


```{r}
#| eval: true

# Final_cohort_first_eye <- Final_cohort %>%
#   group_by(PatientDurableKey) %>%
#   slice_min(Date_retin_dx, n = 1, with_ties = FALSE) %>%
#   ungroup()
#
#
#
#
# # patients with both eyes
```
```

```r
# Final_cohort %>%
#   group_by(PatientDurableKey) %>%
#   summarise(n_eyes = n()) %>%
#   filter(n_eyes == 2) %>%
#   nrow()
#
# # patients with one eye in each group
#  Final_cohort %>%
#   group_by(PatientDurableKey) %>%
#   summarise(n_groups = n_distinct(exposed_group)) %>%
#   filter(n_groups == 2) %>%
#   nrow()


patient_df |>
  # select(age, grade, stage, trt) |>
  # mutate(grade = paste("Grade", grade)) |>
   mutate(
         # hydroxyurea_ever = factor(hydroxyurea_ever, levels = c("0", "1"), labels = c("No hydroxyurea", "hydroxyurea ever")),
         baseline_PDR= factor(baseline_PDR, levels = c("0", "1"), labels = c("No" , "Yes")),
         bilateral_eyes= factor(bilateral_eyes, levels = c("0", "1"), labels = c("No" , "Yes"))
         ) %>%

  tbl_strata(
    strata = exposed_group ,
    .tbl_fun =
      ~ .x |>
          tbl_summary(by=Genotype, include=c(Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
                                             baseline_severity,bilateral_eyes, #baseline_PDR
                                             svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
                                             # Genotype,
                                             Age_retin_dx,age_group,
                                             exposed_at_index,hydroxyurea_ever,
                                             any_progression,days_to_progression,max_followup_months),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(

                        Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
                        race="Race",
                        FU_month="Follow up (months)",
                        insurance='Insurance',
                        baseline_severity='Retinopathy severity',
                          svi_overall_2020 ~ "SVI: Overall",
                  svi_ses_2020 ~ "SVI: Socioeconomic Status",
                  svi_house_2020 ~ "SVI: Housing Composition",
                  svi_minority_2020 ~ "SVI: Minority Status",
                  svi_transport_2020 ~ "SVI: Housing Type / Transportation",
                  exposed_at_index~'Hydroxyurea use at index date',
                  hydroxyurea_ever~'Hydroxyurea ever',
                  any_progression~'Had progression',
                  days_to_progression~'Time to progression (days)'

                        ),
              missing = "ifany"

                )|>
          add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall(),
    .header = "**{strata}**, N = {n}"
  ) %>%
  as_flex_table() %>%
flextable::theme_vanilla()

# patient_df %>%
#   mutate(
#          # hydroxyurea_ever = factor(hydroxyurea_ever, levels = c("0", "1"), labels = c("No hydroxyurea", "hydroxyurea ever")),
#          baseline_PDR= factor(baseline_PDR, levels = c("0", "1"), labels = c("No" , "Yes")),
#          bilateral_eyes= factor(bilateral_eyes, levels = c("0", "1"), labels = c("No" , "Yes"))
#          ) %>%
#   tbl_summary(by=exposed_group, include=c(Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
#                                           baseline_severity,bilateral_eyes, #baseline_PDR
#                                           svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
#                                           Genotype,Age_retin_dx,age_group,
#                                           exposed_at_index,hydroxyurea_ever,
#                                           any_progression,days_to_progression,max_followup_months),
#              type = c(where(is.numeric) ~ "continuous"),
#              statistic = list(
#                all_continuous() ~ "{median} ({p25}, {p75})",
#                all_categorical() ~ "{n} ({p}%)"
#              ),digits = all_continuous() ~ 2,
#              label=list(
#
#                        Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
#                        race="Race",
#                        FU_month="Follow up (months)",
#                        insurance='Insurance',
#                        baseline_severity='Retinopathy severity',
#                          svi_overall_2020 ~ "SVI: Overall",
#                  svi_ses_2020 ~ "SVI: Socioeconomic Status",
#                  svi_house_2020 ~ "SVI: Housing Composition",
#                  svi_minority_2020 ~ "SVI: Minority Status",
#                  svi_transport_2020 ~ "SVI: Housing Type / Transportation",
#                  exposed_at_index~'Hydroxyurea use at index date',
#                  hydroxyurea_ever~'Hydroxyurea ever',
#                  any_progression~'Had progression',
#                  days_to_progression~'Time to progression (days)'
#
#                        ),
#              missing = "ifany"
#
#                ) %>%
#   add_p() %>%
#   bold_p(t = 0.05) %>%
```

```
#   bold_labels() %>%
#   add_overall() %>%
#   modify_header(label = "**Variable**") %>%
#   as_flex_table() %>%
#   # flextable::set_caption("Demographic data of eyes.") %>%
#   # flextable::add_footer_row(values = "Note: 8010 had no baseline IOP value", colwidths = 5) %>%
#   #   flextable::add_footer_row(values = "Note: 'Unknown' under the complication type category are those eyes that had no complications. All other
'Unknown' are true unknowns from the database" , colwidths = 5) %>%
#   flextable::theme_vanilla()




eye_df2 |>
  tbl_strata(
    strata = exposed_group ,
    .tbl_fun =
      ~ .x |>
          tbl_summary(by=Genotype, include=c(Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
                                             severity, #baseline_PDR
                                             svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
                                             Age_retin_dx,
                                             exposed_at_index,
                                             progressed_flag,days_to_progression,
                                           # any_complication,days_complication,
                                           # any_comp_unexposed:any_injection_exposed,event_after_exposure,
                                           # event_only_before_exposure,

                                             FU_month),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(

                        Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
                        race="Race",
                        FU_month="Follow up (months)",
                        insurance='Insurance',
                        severity='Retinopathy severity',
                          svi_overall_2020 ~ "SVI: Overall",
                    svi_ses_2020 ~ "SVI: Socioeconomic Status",
                    svi_house_2020 ~ "SVI: Housing Composition",
                    svi_minority_2020 ~ "SVI: Minority Status",
                    svi_transport_2020 ~ "SVI: Housing Type / Transportation",
                    exposed_at_index~'Hydroxyurea use at index date',
                    # hydroxyurea_ever~'Hydroxyurea ever',
                    progressed_flag~'Had progression',
                    days_to_progression~' Time to progression (days)'
                      ),
              missing = "ifany"

                )|>
          add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall(),
    .header = "**{strata}**, N = {n}"
  ) %>%
  as_flex_table() %>%
flextable::theme_vanilla()



# eye_df2 %>%
#   tbl_summary(by=exposed_group, include=c(Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
#                                           severity, #baseline_PDR
#                                           svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
#                                           Genotype,Age_retin_dx,
#                                           exposed_at_index,
#                                           progressed_flag,days_to_progression,
#                                         # any_complication,days_complication,
#                                         # any_comp_unexposed:any_injection_exposed,event_after_exposure,
#                                         # event_only_before_exposure,
#
#                                           FU_month),
#               type = c(where(is.numeric) ~ "continuous"),
#               statistic = list(
#                 all_continuous() ~ "{median} ({p25}, {p75})",
#                 all_categorical() ~ "{n} ({p}%)"
#               ),digits = all_continuous() ~ 2,
#               label=list(
#
#                         Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
#                         race="Race",
#                         FU_month="Follow up (months)",
#                         insurance='Insurance',
#                         severity='Retinopathy severity',
#                           svi_overall_2020 ~ "SVI: Overall",
#                     svi_ses_2020 ~ "SVI: Socioeconomic Status",
#                     svi_house_2020 ~ "SVI: Housing Composition",
#                     svi_minority_2020 ~ "SVI: Minority Status",
#                     svi_transport_2020 ~ "SVI: Housing Type / Transportation",
#                     exposed_at_index~'Hydroxyurea use at index date',
#                     # hydroxyurea_ever~'Hydroxyurea ever',
#                     progressed_flag~'Had progression',
#                     days_to_progression~' Time to progression (days)'
#
#                     # any_complication~'Had complication',
#                     # days_complication~'Time to complication (days)',
#                     #
#                     # any_comp_unexposed~'Any complication when unexposed',
#                     # any_vh_unexposed~'Complication: vitreous hemorrhage unexposed',
#                     # any_trd_unexposed~'Complication: tractional retinal detachment unexposed',
#                     # any_prp_unexposed~'Complication: prp unexposed',
#                     # any_vitrectomy_unexposed~'Complication: vitrectomy unexposed',
#                     # any_injection_unexposed~'Complication: injection unexposed',
#                     #
```

```
#                   # any_comp_exposed~'Any complication when exposed',
#                   # any_vh_exposed~'Complication: vitreous hemorrhage exposed',
#                   # any_trd_exposed~'Complication: tractional retinal detachment exposed',
#                   # any_prp_exposed~'Complication: prp exposed',
#                   # any_vitrectomy_exposed~'Complication: vitrectomy exposed',
#                   # any_injection_exposed~'Complication: injection exposed',
#                   #
#                   # event_after_exposure~'Had complication after exposure',
#                   # event_only_before_exposure~'Had complication before exposure'
#
#                         ),
#               missing = "ifany"
#
#                 ) %>%
#   add_p() %>%
#   bold_p(t = 0.05) %>%
#   bold_labels() %>%
#   add_overall() %>%
#   modify_header(label = "**Variable**") %>%
#   as_flex_table() %>%
#   # flextable::set_caption("Demographic data of eyes.") %>%
#   # flextable::add_footer_row(values = "Note: 8010 had no baseline IOP value", colwidths = 5) %>%
#   #   flextable::add_footer_row(values = "Note: 'Unknown' under the complication type category are those eyes that had no complications. All other 'Unknown' are true unknowns from the database" , colwidths = 5) %>%
#   flextable::theme_vanilla()









```

Progression

```{r}
#| eval: true


prog_eye_df <- eye_df2 %>%

  # only eyes at risk
  filter(severity == "Nonproliferative") %>%

  mutate(

    # event indicator
    event = ifelse(progressed_flag == "Yes", 1, 0),

    # time to event or censor
    time_days = ifelse(
      event == 1,
      days_to_progression,
      FU_day
    ),

    time_months = time_days / 30,

    # exposure (baseline version for now)
    exposed = ifelse(exposed_at_index == "Yes", 1, 0)
  )

# table(prog_eye_df$event)
# table(prog_eye_df$exposed)
# summary(prog_eye_df$time_years)


library(survival)
library(survminer)

library(ggsurvfit)



survfit2(
    Surv(time_months, event) ~ exposed_group,
    data = prog_eye_df,
    id = PatientDurableKey,
                cluster = PatientDurableKey,
                robust = TRUE,
                na.action = na.omit
  )|>
    ggsurvfit(linetype_aes = TRUE, linewidth = 1) +
    labs(
      title = "Kaplan-Meier curve for progression",
      x = "Time since retinopathy diagnosis (Month)",
      y = "Survival probability",
      color = NULL,
      linetype = NULL
    ) +
    add_confidence_interval() +
    add_censor_mark() +
    add_pvalue() +
  # annotate("text", x = 8, y = 0.55,
  #         label = "Clustered cox model p = 0.4")+
    add_risktable(
      risktable_stats = c("n.risk", "n.event", "n.censor"),
      stats_label = c("At Risk", "Events", "Censored")
    ) +
    scale_color_manual(values = c("black", "black")) +
    scale_fill_grey() +
    scale_y_continuous(limits = c(0.5, 1)) +
    # scale_x_continuous(
    #   limits = c(0, 5),
    #   # breaks = seq(0, 12, 2)
    # ) +
    theme(
      panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
```

```
    panel.background = element_blank()
  )



```
Cox ph model for progression

In the unadjusted analysis, hydroxyurea exposure was associated with a significantly lower risk of progression (HR 0.45, 95% CI 0.24–0.83, p=0.011). However, after adjustment for demographic and clinical factors, including genotype, this association was attenuated and no longer statistically significant.

In the time-varying Cox proportional hazards model accounting for changes in hydroxyurea exposure status over follow-up and inter-eye correlation, hydroxyurea exposure was not significantly associated with risk of retinopathy progression (hazard ratio [HR] 0.88, 95% CI 0.40–1.96, p=0.76). Older age at retinopathy diagnosis was independently associated with increased risk of progression (HR 1.02 per year, 95% CI 1.01–1.03, p=0.001). Sex was not significantly associated with progression risk. Compared with the reference genotype group, Hb-SB, Hb-SC, and Hb-SS genotypes were associated with significantly lower hazards of progression.

No evidence that hydroxyurea reduces progression, but results are imprecise (likely underpowered)

```{r}
#| eval: true




#
surv_data <- prog_eye_df %>%
  select(PatientDurableKey,time_months,event,exposed_group,age=Age_retin_dx,Sex,Genotype,race=FirstRace) %>% #,region,insurance
  filter(Genotype!='Other sickle-cell disorders') %>% droplevels()



surv_data$exposed_group <- relevel(surv_data$exposed_group, ref='unexposed')



# cox_model <- coxph(
#   Surv(time_months, event) ~ exposed_group + age + Sex + Genotype+race ,
#   data = surv_data,
#   cluster = PatientDurableKey
# )
#
# # summary(cox_model)
#
# coxph <- cox.zph((cox_model))
#
# plot(coxph)
#
#
# surv_data %>% tbl_summary(by=exposed_group,
#                           include=c(-PatientDurableKey),
#                           label=list(
#
#                           time_months="Time to progression (months)",
#                           event="Had progression"
#
#
#                           )
#                           ) %>%
#   add_p() %>%
#   bold_p(t = 0.05) %>%
#   bold_labels() %>%
#   add_overall() %>%
#   modify_header(label = "**Variable**") %>%
#   as_flex_table() %>%
#   flextable::theme_vanilla()
#
#
# tbl_regression(cox_model, exponentiate = T)
#
#
# # names(prog_eye_df)
# # prog_eye_df$Genotype
# #
# # table(prog_eye_df$Genotype)
#
#
#
# simple_mod <- coxph(Surv(time_months, event) ~ exposed_group,
#   data = surv_data,
#   cluster = PatientDurableKey)
#
# tbl_regression(simple_mod, exponentiate = T)




#---time varying

# prog_eye_df <- prog_eye_df %>%
#   mutate(
#     exposed_tv =
#       ifelse(!is.na(exposure_start_date) &
#              time_days >= as.numeric(exposure_start_date - Date_retin_dx),
#              1, 0)
#   )
#
#
#
# tv_data <- eye_df2 %>%
#
#   # only NPDR eyes (at risk)
#   filter(severity == "Nonproliferative") %>%
#
#   mutate(
```

```
#
#     event = ifelse(progressed_flag == "Yes", 1, 0),
#
#     # total follow-up time
#     tstop_total = ifelse(event == 1, days_to_progression, FU_day),
#
#     # exposure start relative to index
#     tstart_exposure = as.numeric(exposure_start_date - Date_retin_dx)
#   )
#
#
#
#
# tv_long <- tv_data %>%
#
#   rowwise() %>%
#
#   do({
#
#     row <- .
#
#     # no exposure ever
#     if (is.na(row$tstart_exposure)) {
#
#       data.frame(
#         PatientDurableKey = row$PatientDurableKey,
#         UID = row$UID,
#
#         tstart = 0,
#         tstop = row$tstop_total,
#
#         event = row$event,
#         exposed = 0,
#
#         age = row$Age_retin_dx,
#         Sex = row$Sex,
#         race = row$FirstRace,
#         Genotype = row$Genotype
#       )
#
#     } else {
#
#       # exposure happens AFTER follow-up → treat as unexposed
#       if (row$tstart_exposure >= row$tstop_total) {
#
#         data.frame(
#           PatientDurableKey = row$PatientDurableKey,
#           UID = row$UID,
#
#           tstart = 0,
#           tstop = row$tstop_total,
#
#           event = row$event,
#           exposed = 0,
#
#           age = row$Age_retin_dx,
#           Sex = row$Sex,
#           race = row$FirstRace,
#           Genotype = row$Genotype
#         )
#
#       } else {
#
#         # TRUE time-varying case
#
#         rbind(
#
#           # interval 1: unexposed
#           data.frame(
#             PatientDurableKey = row$PatientDurableKey,
#             UID = row$UID,
#
#             tstart = 0,
#             tstop = row$tstart_exposure,
#
#             event = 0,
#             exposed = 0,
#
#             age = row$Age_retin_dx,
#             Sex = row$Sex,
#             race = row$FirstRace,
#             Genotype = row$Genotype
#           ),
#
#           # interval 2: exposed
#           data.frame(
#             PatientDurableKey = row$PatientDurableKey,
#             UID = row$UID,
#
#             tstart = row$tstart_exposure,
#             tstop = row$tstop_total,
#
#             event = row$event,
#             exposed = 1,
#
#             age = row$Age_retin_dx,
#             Sex = row$Sex,
#             race = row$FirstRace,
#             Genotype = row$Genotype
#           )
#         )
#       }
#     }
#
#   }) %>%
#
#   ungroup()
#
#
#
```

```
# cox_tv <- coxph(
#   Surv(tstart, tstop, event) ~ exposed + age + Sex + Genotype+ race,
#   data = tv_long,
#   cluster = PatientDurableKey
# )
#
# summary(cox_tv)
#
# tbl_regression(cox_tv, exponentiate = T)


# ---- version 2

surv_eye <- eye_df2 %>%

  filter(severity == "Nonproliferative") %>%

  mutate(

    event =
      ifelse(progressed_flag == "Yes", 1, 0),

    eventtime =
      ifelse(
        event == 1,
        days_to_progression,
        FU_day
      )
  ) %>%

  select(
    UID,
    PatientDurableKey,
    eventtime,
    event,
    Age_retin_dx,
    Sex,
    Genotype,
    exposure_start_date,
    Date_retin_dx
  )

# Create delayed exposure variable

surv_eye <- surv_eye %>%

  mutate(

    hydroxy_start_lag =
      pmax(
        as.numeric(exposure_start_date - Date_retin_dx) + 90,
        0
      )
  )

# min(surv_eye$hydroxy_start_lag, na.rm = T)

# testdf <- eye_df2 %>% group_by(UID) %>% filter(n()>1) %>% ungroup()


tv_dat <- tmerge(
  data1 = surv_eye,
  data2 = surv_eye,

  id = UID,

  event = event(eventtime, event)
)

tv_dat <- tmerge(
  tv_dat,
  surv_eye,

  id = UID,

  hydroxy_exposure =
    tdc(hydroxy_start_lag)
)

cox_tv <- coxph(

  Surv(tstart, tstop, event) ~
    hydroxy_exposure +
    Age_retin_dx +
    Sex +
    Genotype,

  cluster = PatientDurableKey,

  data = tv_dat
)

summary(cox_tv)

tbl_regression(cox_tv, exponentiate = T)

```

Complications

In a time-varying Cox proportional hazards model accounting for delayed hydroxyurea initiation and inter-eye correlation, hydroxyurea exposure was associated with a 57% lower hazard of severe retinal complications (HR 0.43, 95% CI 0.24–0.76, p=0.004).

after accounting for:

delayed treatment initiation
inter-eye correlation

genotype
demographics

```{r}
#| eval: true

eye_df2 %>% mutate(time_to_severe=ifelse(!is.na(first_severe_type),time_to_severe,NA),
                   hydroxyurea_exposed_event=ifelse(hydroxyurea_exposed_event==0,'No','Yes'),
                   severe_event=ifelse(severe_event==0,'No','Yes')

                   ) %>%
  tbl_summary(by=exposed_group, include=c(
                                          # Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
                                          # severity, #baseline_PDR
                                          # svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
                                          # Genotype,Age_retin_dx,
                                          # exposed_at_index,
                                          # progressed_flag,days_to_progression,
                                        first_severe_type:time_to_severe

                                          ),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(

                  #        Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
                  #        race="Race",
                  #        FU_month="Follow up (months)",
                  #        insurance='Insurance',
                  #        severity='Retinopathy severity',
                  #          svi_overall_2020 ~ "SVI: Overall",
                  # svi_ses_2020 ~ "SVI: Socioeconomic Status",
                  # svi_house_2020 ~ "SVI: Housing Composition",
                  # svi_minority_2020 ~ "SVI: Minority Status",
                  # svi_transport_2020 ~ "SVI: Housing Type / Transportation",
                  # exposed_at_index~'Hydroxyurea use at index date',
                  # # hydroxyurea_ever~'Hydroxyurea ever',
                  # progressed_flag~'Had progression',

                  time_to_severe~'Time to complication (days)',
                  first_severe_type~'Complication type',
                  severe_event~'Total eyes that had a complication',
                  hydroxyurea_exposed_event~'Total complication after hydroxyurea use ( + 90 days)'

                        ),
              missing = "no"

                ) %>%
  add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall() %>%
  modify_header(label = "**Variable**") %>%
  as_flex_table() %>%
  # flextable::set_caption("Demographic data of eyes.") %>%
  # flextable::add_footer_row(values = "Note: 8010 had no baseline IOP value", colwidths = 5) %>%
  #   flextable::add_footer_row(values = "Note: 'Unknown' under the complication type category are those eyes that had no complications. All other
'Unknown' are true unknowns from the database" , colwidths = 5) %>%
  flextable::theme_vanilla()


eye_df2 %>% mutate(time_to_severe=ifelse(!is.na(first_severe_type),time_to_severe,NA),
                   hydroxyurea_exposed_event=ifelse(hydroxyurea_exposed_event==0,'No','Yes'),
                   severe_event=ifelse(severe_event==0,'No','Yes')
                   ) %>%
  tbl_strata(
    strata = exposed_group ,
    .tbl_fun =
      ~ .x |>
          tbl_summary(by=Genotype, include=c(
                                          first_severe_type:time_to_severe
                                            ),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(
                  time_to_severe~'Time to complication (days)',
                  first_severe_type~'Complication type',
                  severe_event~'Total eyes that had a complication',
                  hydroxyurea_exposed_event~'Total complication after hydroxyurea use ( + 90 days)'

                        ),
              missing = "no"

                )|>
          add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall(),
    .header = "**{strata}**, N = {n}"
  ) %>%
  as_flex_table() %>%
flextable::theme_vanilla()




# names(first_complication_by_exposure)
```

```r
# names(eye_df2)


comp_surv <- eye_df2 %>%

  dplyr::select(
    UID,
    PatientDurableKey,
    Date_retin_dx,
    FU_day,
    exposure_start_date,
    Age_retin_dx,
    Sex,
    Genotype,
    Date_complication=first_severe_date,
    complication_type=first_severe_type
  )


comp_surv <- comp_surv %>%

  mutate(

    severe_event =
      ifelse(
        !is.na(Date_complication),
        1,
        0
      ),

    eventtime =
      ifelse(
        severe_event == 1,

        as.numeric(
          Date_complication - Date_retin_dx
        ),

        FU_day
      )
  )

comp_surv <- comp_surv %>%

  mutate(

    hydroxy_start_lag =
      pmax(
        as.numeric(
          exposure_start_date - Date_retin_dx
        ) + 90,
        0
      )
  )


tv_comp <- tmerge(

  data1 = comp_surv,
  data2 = comp_surv,

  id = UID,
  tstop = eventtime,
  severe_event =
    event(eventtime, severe_event)
)


tv_comp <- tmerge(

  tv_comp,
  comp_surv,

  id = UID,

  hydroxy_exposure =
    tdc(hydroxy_start_lag)
)


cox_comp <- coxph(

  Surv(
    tstart,
    tstop,
    severe_event
  ) ~

    hydroxy_exposure +
    Age_retin_dx +
    Sex +
    Genotype,

  cluster = PatientDurableKey,

  data = tv_comp
)

summary(cox_comp)



tbl_regression(cox_comp, exponentiate = T)


# cox.zph(cox_comp)
#
# plot(cox.zph(cox_comp))
```

```
#-------- interaction

# table(tv_comp$genotype_collapse)

tv_comp <- tv_comp %>%
  mutate(genotype_collapse=case_when
         (Genotype=='Hb-SS'~Genotype,
          TRUE~  'Non Hb-SS'
           ),
         hydroxy_exposure_yn=factor(ifelse(hydroxy_exposure==0,'No','Yes'))
         )

cox_int <- coxph(
  Surv(tstart, tstop, severe_event) ~
    hydroxy_exposure_yn * genotype_collapse +
    Age_retin_dx +
    Sex,
  data = tv_comp,
  cluster = PatientDurableKey
)

drop1(cox_int, test = "Chisq")



summary(cox_int)

tbl_regression(cox_int, exponentiate = T)


library(emmeans)

emm <- emmeans(
  cox_int,
  ~ hydroxy_exposure_yn | genotype_collapse
)

library(gt)

summary(
  pairs(emm, reverse = TRUE),
  infer = c(TRUE, TRUE),
  type = "response"
) %>%  as.data.frame() %>%
  mutate(
    HR_CI = sprintf(
      "%.2f (%.2f, %.2f)",
      ratio, asymp.LCL , asymp.UCL
    ),
    p.value = scales::pvalue(p.value)


  ) %>%
  select(
    Genotype = genotype_collapse,
    Exposure = contrast,
    `Hazard Ratio (95% CI)` = HR_CI,
    `p-value` = p.value
  ) %>%
  gt() %>%
  tab_header(
    title = "Estimated marginal hazard ratios by genotype"
  )


table(tv_comp$hydroxy_exposure )




tv_comp$hydroxy_exposure <- factor(
  tv_comp$hydroxy_exposure,
  levels = c(0,1),
  labels = c("Unexposed","Hydroxyurea")
)


# sm_fit <- survfit(
#   Surv(tstart, tstop, severe_event) ~ hydroxy_exposure,
#   data = tv_comp
# )




# ggsurvplot(
#   sm_fit,
#
#   data = tv_comp,
#
#   conf.int = TRUE,
#
#   risk.table = TRUE,
#
#   censor = FALSE,
#
#   xlab = "Days from Initial Sickle Cell Retinopathy Diagnosis",
#
#   ylab = "Complication-Free Survival",
#
#   legend.title = "",
#
#   legend.labs = c(
#     "Unexposed",
#     "Hydroxyurea"
#   ),
#
#   # ggtheme = theme_bw()
```

```
#    ggtheme = ggpubr::theme_pubr()
# )


survfit2(

  Surv(
    tstart,
    tstop,
    severe_event
  ) ~ hydroxy_exposure,

  data = tv_comp %>% mutate(
    tstart = tstart / 30.4,
    tstop = tstop / 30.4
  )
  ,

  id = UID,

  cluster = PatientDurableKey,

  robust = TRUE,

  na.action = na.omit

) |>

  ggsurvfit(
    linetype_aes = TRUE,
    linewidth = 1
  ) +

  labs(
    title = "Simon-Makuch curve for retinal complications",

    x = "Time since retinopathy diagnosis (Months)",

    y = "Complication-free survival probability",

    color = NULL,

    linetype = NULL
  ) +

  add_confidence_interval() +

  add_censor_mark() +

  add_risktable(
    risktable_stats = c(
      "n.risk",
      "n.event",
      "n.censor"
    ),

    stats_label = c(
      "At Risk",
      "Events",
      "Censored"
    )
  ) +

  annotate(
    "text",

    x = 24,

    y = 0.75,

    label =
      "Time-varying Cox HR 0.45 (95% CI 0.25–0.8), p = 0.007"
  ) +

  scale_color_manual(
    values = c("black", "black")
  ) +

  scale_fill_grey() +

  scale_y_continuous(
    limits = c(0.5, 1)
  ) +

  theme(
    panel.grid.major = element_blank(),

    panel.grid.minor = element_blank(),

    panel.background = element_blank()
  )



## two plots for stratified.



tv_comp_hbss <- tv_comp %>% filter(genotype_collapse == "Hb-SS") %>% droplevels() %>%  mutate(
    tstart = tstart / 30.4,
    tstop = tstop / 30.4
)

tv_comp_non_hbss <- tv_comp %>% filter(genotype_collapse == 'Non Hb-SS') %>% droplevels() %>%  mutate(
    tstart = tstart / 30.4,
    tstop = tstop / 30.4
)
```

```r
survfit2(

  Surv(
    tstart,
    tstop,
    severe_event
  ) ~ hydroxy_exposure,

  data = tv_comp_hbss,

  id = UID,

  cluster = PatientDurableKey,

  robust = TRUE,

  na.action = na.omit

) |>

  ggsurvfit(
    linetype_aes = TRUE,
    linewidth = 1
  ) +

  labs(
    title = "Simon-Makuch curve for retinal complications: Hb-SS",

    x = "Time since retinopathy diagnosis (Months)",

    y = "Complication-free survival probability",

    color = NULL,

    linetype = NULL
  ) +

  add_confidence_interval() +

  add_censor_mark() +

  add_risktable(
    risktable_stats = c(
      "n.risk",
      "n.event",
      "n.censor"
    ),

    stats_label = c(
      "At Risk",
      "Events",
      "Censored"
    )
  ) +

  annotate(
    "text",

    x = 24,

    y = 0.75,

    label =
      "Time-varying Cox HR 0.25 (95% CI 0.12-0.51), p = <0.001"
  ) +

  scale_color_manual(
    values = c("black", "black")
  ) +

  scale_fill_grey() +

  scale_y_continuous(
    limits = c(0.5, 1)
  ) +

  theme(
    panel.grid.major = element_blank(),

    panel.grid.minor = element_blank(),

    panel.background = element_blank()
  )




survfit2(

  Surv(
    tstart,
    tstop,
    severe_event
  ) ~ hydroxy_exposure,

  data = tv_comp_non_hbss,

  id = UID,

  cluster = PatientDurableKey,

  robust = TRUE,

  na.action = na.omit

) |>

  ggsurvfit(
    linetype_aes = TRUE,
```

```
    linewidth = 1
  ) +

  labs(
    title = "Simon-Makuch curve for retinal complications: Non Hb-SS",

    x = "Time since retinopathy diagnosis (Months)",

    y = "Complication-free survival probability",

    color = NULL,

    linetype = NULL
  ) +

  add_confidence_interval() +

  add_censor_mark() +

  add_risktable(
    risktable_stats = c(
      "n.risk",
      "n.event",
      "n.censor"
    ),

    stats_label = c(
      "At Risk",
      "Events",
      "Censored"
    )
  ) +

  annotate(
    "text",

    x = 24,

    y = 0.75,

    label =
      "Time-varying Cox HR 0.1.88 (95% CI 0.78-0.4.56), p = 0.162"
  ) +

  scale_color_manual(
    values = c("black", "black")
  ) +

  scale_fill_grey() +

  scale_y_continuous(
    limits = c(0.5, 1)
  ) +

  theme(
    panel.grid.major = element_blank(),

    panel.grid.minor = element_blank(),

    panel.background = element_blank()
  )



```
Complication burden

descriptive stats

```{r}
#| eval: true

#
eye_df %>%
  tbl_summary(by=exposed_group, include=c(
                                          # Sex,FirstRace,Ethnicity,Language, insurance,#StateOrProvince,
                                          # severity, #baseline_PDR
                                          # svi_overall_2020,svi_ses_2020,svi_house_2020,svi_minority_2020,svi_transport_2020,ruca,region,
                                          # Genotype,Age_retin_dx,
                                          # exposed_at_index,
                                          # progressed_flag,days_to_progression,
                                        any_complication,days_complication,
                                        any_comp_unexposed:any_injection_exposed,event_after_exposure,
                                        event_only_before_exposure

                                          ),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(

                  #        Age_retin_dx="Age at first sickle cell retinopathy diagnosis (years)",
                  #        race="Race",
                  #        FU_month="Follow up (months)",
                  #        insurance='Insurance',
                  #        severity='Retinopathy severity',
                  #          svi_overall_2020 ~ "SVI: Overall",
                  # svi_ses_2020 ~ "SVI: Socioeconomic Status",
                  # svi_house_2020 ~ "SVI: Housing Composition",
                  # svi_minority_2020 ~ "SVI: Minority Status",
                  # svi_transport_2020 ~ "SVI: Housing Type / Transportation",
                  # exposed_at_index~'Hydroxyurea use at index date',
                  # # hydroxyurea_ever~'Hydroxyurea ever',
                  # progressed_flag~'Had progression',

                  any_complication~'Had complication',
                  days_complication~'Time to complication (days)',
```

```
                  any_comp_unexposed~'Any complication when unexposed',
                  any_vh_unexposed~'Complication: vitreous hemorrhage unexposed',
                  any_trd_unexposed~'Complication: tractional retinal detachment unexposed',
                  any_prp_unexposed~'Complication: prp unexposed',
                  any_vitrectomy_unexposed~'Complication: vitrectomy unexposed',
                  any_injection_unexposed~'Complication: injection unexposed',

                  any_comp_exposed~'Any complication when exposed',
                  any_vh_exposed~'Complication: vitreous hemorrhage exposed',
                  any_trd_exposed~'Complication: tractional retinal detachment exposed',
                  any_prp_exposed~'Complication: prp exposed',
                  any_vitrectomy_exposed~'Complication: vitrectomy exposed',
                  any_injection_exposed~'Complication: injection exposed',

                  event_after_exposure~'Had complication after exposure',
                  event_only_before_exposure~'Had complication before exposure'

                        ),
              missing = "no"

                ) %>%
  add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall() %>%
  modify_header(label = "**Variable**") %>%
  as_flex_table() %>%
  # flextable::set_caption("Demographic data of eyes.") %>%
  # flextable::add_footer_row(values = "Note: 8010 had no baseline IOP value", colwidths = 5) %>%
  #   flextable::add_footer_row(values = "Note: 'Unknown' under the complication type category are those eyes that had no complications. All other
'Unknown' are true unknowns from the database" , colwidths = 5) %>%
  flextable::theme_vanilla




eye_df |>
  tbl_strata(
    strata = exposed_group ,
    .tbl_fun =
      ~ .x |>
          tbl_summary(by=Genotype, include=c(
                                          any_complication,days_complication,
                                          any_comp_unexposed:any_injection_exposed,event_after_exposure,
                                          event_only_before_exposure

                                            ),
              type = c(where(is.numeric) ~ "continuous"),
              statistic = list(
                all_continuous() ~ "{median} ({p25}, {p75})",
                all_categorical() ~ "{n} ({p}%)"
              ),digits = all_continuous() ~ 2,
              label=list(

                  any_complication~'Had complication',
                  days_complication~'Time to complication (days)',

                  any_comp_unexposed~'Any complication when unexposed',
                  any_vh_unexposed~'Complication: vitreous hemorrhage unexposed',
                  any_trd_unexposed~'Complication: tractional retinal detachment unexposed',
                  any_prp_unexposed~'Complication: prp unexposed',
                  any_vitrectomy_unexposed~'Complication: vitrectomy unexposed',
                  any_injection_unexposed~'Complication: injection unexposed',

                  any_comp_exposed~'Any complication when exposed',
                  any_vh_exposed~'Complication: vitreous hemorrhage exposed',
                  any_trd_exposed~'Complication: tractional retinal detachment exposed',
                  any_prp_exposed~'Complication: prp exposed',
                  any_vitrectomy_exposed~'Complication: vitrectomy exposed',
                  any_injection_exposed~'Complication: injection exposed',

                  event_after_exposure~'Had complication after exposure',
                  event_only_before_exposure~'Had complication before exposure'

                        ),
              missing = "no"

                )|>
          add_p() %>%
  bold_p(t = 0.05) %>%
  bold_labels() %>%
  add_overall(),
    .header = "**{strata}**, N = {n}"
  ) %>%
  as_flex_table() %>%
flextable::theme_vanilla()







complication_boxplt

```

GEE model

```{r}
#| eval: true

tbl_regression(
```

```
  gee_comp,

  exponentiate = TRUE,

  label = list(

    exposed_group ~ "Hydroxyurea exposure",

    Age_retin_dx ~ "Age at retinopathy diagnosis",

    Sex ~ "Sex",

    Genotype ~ "Genotype"
  )

) %>%

  bold_labels() %>%

  italicize_levels()

```

VA

```{r, VA}
#| eval: true

va_long <- va_timepoints %>%
  mutate(UID=paste0(PatientDurableKey, "_", Laterality)) %>%

  pivot_longer(

    cols = c(
      baseline_va,
      va_3m,
      va_6m,
      va_9m,
      va_1y,
      va_2y,
      va_3y,
      va_5y
    ),

    names_to = "timepoint",

    values_to = "VA"
  )%>%

  mutate(

     timepoint = factor(
      timepoint,
      levels = c(
        "baseline_va",
        "va_3m",
        "va_6m",
        "va_9m",
        "va_1y",
        "va_2y",
        "va_3y",
        "va_5y"
      ),
      labels = c(
        "Baseline",
        "3 Months",
        "6 Months",
        "9 Months",
        "1 Year",
        "2 Years",
        "3 Years",
        "5 Years"
      ),
      ordered = TRUE
    )
  ) %>% filter(!is.na(VA)) %>%
  left_join(

    eye_df2 %>%
      select(
        UID,
        exposed_group,
        Age_retin_dx,
        Sex,
        Genotype,
        progressed_flag,
        severity
      ) %>%
      distinct(),

    by = "UID"
  )


# table(va_long$severity)


#
va_long %>%

  group_by(
    exposed_group,
    timepoint
  ) %>%

  summarise(

    n = sum(!is.na(VA)),
```

```
    mean_va = mean(VA, na.rm = TRUE),

    sd_va = sd(VA, na.rm = TRUE),

    median_va = median(VA, na.rm = TRUE),

    iqr_va = IQR(VA, na.rm = TRUE),

    .groups = "drop"
  )%>%
  gt(groupname_col = "exposed_group") %>%
  fmt_number(
    columns = c(mean_va, sd_va, median_va, iqr_va),
    decimals = 2
  ) %>%
  cols_label(
    timepoint = "Timepoint",
    n = "N",
    mean_va = "Mean VA",
    sd_va = "SD",
    median_va = "Median VA",
    iqr_va = "IQR"
  ) %>%
  tab_header(
    title = "Visual Acuity Summary Statistics"
  )



# Counts per group/timepoint
va_counts <- va_long %>%
  group_by(exposed_group, timepoint) %>%
  summarise(
    n = sum(!is.na(VA)),
    .groups = "drop"
  )

ggplot(
  va_long,
  aes(
    x = timepoint,
    y = VA,
    fill = exposed_group
  )
) +

  # Distribution of VA over time
  geom_boxplot(
    alpha = 0.6,
    outlier.alpha = 0.3,
    position = position_dodge(width = 0.8),
    width = 0.7
  ) +

  # Median points
  # stat_summary(
  #   aes(color = exposed_group),
  #   fun = median,
  #   geom = "point",
  #   position = position_dodge(width = 0.8),
  #   size = 2
  # ) +

  # Add sample sizes above boxes
  geom_text(
    data = va_counts,
    aes(
      x = factor(timepoint),
      y = max(va_long$VA, na.rm = TRUE) + 0.1,
      label = paste0("n=", n),
      group = exposed_group
    ),
    position = position_dodge(width = 0.8),
    size = 3,
    inherit.aes = FALSE
  ) +

  labs(
    title = "Longitudinal Visual Acuity Distributions",
    x = "Time from retinopathy diagnosis",
    y = "Visual acuity (LogMAR)",
    fill = "Hydroxyurea Exposure",
    color = NULL
  ) +

  theme_bw() +

  theme(
    legend.position = "top"
  )









##########################

# str(va_long)

va_long_progression <- va_long %>% filter(severity=='Nonproliferative') %>% droplevels()
```

```
va_counts2 <- va_long_progression%>%
  group_by(
    progressed_flag,
    exposed_group,
    timepoint
  ) %>%
  summarise(
    n = sum(!is.na(VA)),
    .groups = "drop"
  )



ggplot(
  va_long_progression,
  aes(
    x = timepoint,
    y = VA,
    fill = exposed_group
  )
) +

  geom_boxplot(
    alpha = 0.7,
    outlier.alpha = 0.25,
    position = position_dodge(width = 0.8)
  ) +

  facet_wrap(
    ~ progressed_flag,
    nrow = 1
  ) +

  labs(
    title = "Visual Acuity Over Time by Medication Exposure and Retinopathy Progression",
    x = "Follow-up Time",
    y = "Visual Acuity (LogMAR)",
    fill = "Hydroxyurea Exposure"
  ) +

  # Add sample sizes above boxes
  geom_text(
    data = va_counts2,
    aes(
      x = factor(timepoint),
      y = max(va_long$VA, na.rm = TRUE) + 0.1,
      label = paste0("n=", n),
      group = exposed_group
    ),
    position = position_dodge(width = 0.8),
    size = 3,
    inherit.aes = FALSE
  ) +

  theme_bw() +

  theme(
    legend.position = "top",
    strip.background = element_rect(fill = "grey95"),
    strip.text = element_text(face = "bold")
  )
```

VA modeling

A linear mixed-effects model with nested random intercepts for patient and eye was used to evaluate longitudinal changes in visual acuity (VA) between exposed and unexposed groups while accounting for inter-eye correlation and repeated measurements within eyes.

At baseline, unexposed eyes demonstrated significantly worse VA compared with exposed eyes (β = −0.035 logMAR, SE = 0.016, p = 0.031). Increasing age at retinopathy diagnosis was independently associated with worse VA over time (β = 0.003 logMAR per year, SE = 0.0003, p < 0.001). Sex and genotype were not significantly associated with longitudinal VA outcomes.

Among exposed eyes, no significant changes in VA from baseline were observed at 3 months, 6 months, 9 months, 1 year, or ≥2 years. Similarly, within the unexposed group, no significant longitudinal VA changes from baseline were identified after adjustment for multiple comparisons.

A significant interaction between exposure status and the 6-month timepoint was observed (β = 0.059, SE = 0.027, p = 0.032), suggesting differential VA trajectories between groups at 6 months. However, adjusted marginal mean comparisons between exposed and unexposed groups at individual follow-up timepoints were not statistically significant after Bonferroni correction.

Estimated marginal means demonstrated comparable adjusted VA trajectories between groups across longitudinal follow-up.

```{r}
#| eval: true



# table(va_long$timepoint)

va_long_model <- va_long %>%

  mutate(

    time_group = case_when(

      timepoint %in% c("2 Years", "3 Years","5 Years") ~ ">= 2 Years",
      TRUE ~ timepoint
    ),

     time_group = factor(
      time_group,
      levels = c(
        "Baseline",
        "3 Months",
        "6 Months",
        "9 Months",
        "1 Year",
        ">= 2 Years"
```

```
      )
    )

  )

lmm_va <- lmer(

  VA ~
    time_group *
    exposed_group +
    Age_retin_dx +
    Sex +
    Genotype +
    (1 | PatientDurableKey/UID),

  data = va_long_model
)

summary(lmm_va)
emm

# tbl_regression(lmm_va, exponentiate = TRUE)



emm <- emmeans(

  lmm_va,

  ~ time_group * exposed_group
)

# contrast(
#   emm,
#   method = "revpairwise",
#   by = "time_group",
#   adjust = "bonferroni"
# )

emm_time <- emmeans(
  lmm_va,
  ~ time_group | exposed_group
)

# pairs(
#   emm_time,
#   adjust = "bonferroni"
# )

# contrast(
#   emm_time,
#   method = "trt.vs.ctrl",
#   ref = "Baseline"
# )



library(gtsummary)
library(broom.mixed)




  tbl_regression(
    lmm_va,
    intercept = TRUE,
    estimate_fun = ~style_sigfig(.x, digits = 3)
  ) %>%
  bold_labels()

  # tbl_model <-
  # tbl_regression(
  #   lmm_va,
  #
  #   intercept = FALSE,
  #
  #   label = list(
  #     time_group ~ "Follow-up Time",
  #     exposed_group ~ "Medication Exposure",
  #     Age_retin_dx ~ "Age at Retinopathy Diagnosis",
  #     Sex ~ "Sex",
  #     Genotype ~ "Genotype",
  #
  #     `time_group3 Months` ~ "3 Months vs Baseline",
  #     `time_group6 Months` ~ "6 Months vs Baseline",
  #     `time_group9 Months` ~ "9 Months vs Baseline",
  #     `time_group1 Year` ~ "1 Year vs Baseline",
  #     `time_group>= 2 Years` ~ "≥2 Years vs Baseline",
  #
  #     `exposed_groupunexposed` ~ "Unexposed vs Exposed",
  #
  #     `time_group3 Months:exposed_groupunexposed` ~
  #       "3 Months × Unexposed",
  #
  #     `time_group6 Months:exposed_groupunexposed` ~
  #       "6 Months × Unexposed",
  #
  #     `time_group9 Months:exposed_groupunexposed` ~
  #       "9 Months × Unexposed",
  #
  #     `time_group1 Year:exposed_groupunexposed` ~
  #       "1 Year × Unexposed",
  #
  #     `time_group>= 2 Years:exposed_groupunexposed` ~
  #       "≥2 Years × Unexposed"
  #   ),
  #
  #   estimate_fun = ~style_sigfig(.x, digits = 3)
  #
```

```
  # ) %>%
  #
  # bold_p(t = 0.05) %>%
  #
  # modify_header(
  #   estimate ~ "**β**",
  #   ci ~ "**95% CI**",
  #   p.value ~ "**P value**"
  # ) %>%
  #
  # italicize_levels()









  emm_between <- contrast(
  emm,
  method = "revpairwise",
  by = "time_group",
  adjust = "bonferroni"
)

between_df <-
  summary(emm_between, infer = TRUE) %>%
  as.data.frame()


  between_df %>%

  select(
    time_group,
    contrast,
    estimate,
    SE,
    lower.CL=asymp.LCL,
    upper.CL=asymp.UCL,
    p.value
  ) %>%

  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    lower.CL = round(lower.CL, 3),
    upper.CL = round(upper.CL, 3),
    p.value = round(p.value, 3)
  ) %>%

  gt() %>%

  cols_label(
    time_group = "Timepoint",
    contrast = "Comparison",
    estimate = "Adjusted Mean Difference",
    SE = "SE",
    lower.CL = "Lower 95% CI",
    upper.CL = "Upper 95% CI",
    p.value = "P value"
  ) %>%

  tab_header(
    title = md("**Adjusted VA Comparisons Between Exposure Groups**"),
    subtitle = "Estimated marginal means from linear mixed-effects model"
  ) %>%

  fmt_number(
    columns = c(estimate, SE, lower.CL, upper.CL),
    decimals = 3
  ) %>%

  fmt_number(
    columns = p.value,
    decimals = 3
  )


  within_df <-
  contrast(
    emm_time,
    method = "trt.vs.ctrl",
    ref = "Baseline"
  ) %>%

  summary(infer = TRUE) %>%
  as.data.frame()

 within_df %>%

  select(
    exposed_group,
    contrast,
    estimate,
    SE,
 lower.CL=asymp.LCL,
    upper.CL=asymp.UCL,
    p.value
  ) %>%

  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    lower.CL = round(lower.CL, 3),
    upper.CL = round(upper.CL, 3),
    p.value = round(p.value, 3)
  ) %>%
```

```
  gt(groupname_col = "exposed_group") %>%

  cols_label(
    contrast = "Comparison",
    estimate = "Adjusted Mean Change",
    SE = "SE",
    lower.CL = "Lower 95% CI",
    upper.CL = "Upper 95% CI",
    p.value = "P value"
  ) %>%

  tab_header(
    title = md("**Within-Group Longitudinal Changes in VA**"),
    subtitle = "Change from baseline within each exposure group"
  ) %>%

  fmt_number(
    columns = c(estimate, SE, lower.CL, upper.CL),
    decimals = 3
  ) %>%

  fmt_number(
    columns = p.value,
    decimals = 3
  )


# emm_df <- as.data.frame(emm)
#
# library(ggplot2)
#
# ggplot(
#   emm_df,
#   aes(
#     x = time_group,
#     y = emmean,
#     color = exposed_group,
#     group = exposed_group
#   )
# ) +
#   geom_line() +
#   geom_point() +
#   geom_errorbar(
#     aes(ymin = asymp.LCL, ymax = asymp.UCL),
#     width = 0.1
#   ) +
#   labs(
#     y = "Adjusted VA (logMAR)",
#     x = "Time"
#   ) +
#   theme_bw()

```

Revisions

For the complication types, would it be possible to provide a breakdown by genotype and hydroxyurea exposure status? Specifically, it would be helpful to see rates of each complication type in HbAS compared with HbSS, HbSC, and HbSβ-thalassemia to confirm whether the genotype patterns track with expected clinical behavior.

```{r}
# Tue Jun 23 08:55:51 2026 ------------------------------


denom <- eye_df2 %>%
  distinct(
    UID,
    Genotype,
    exposed_group
  ) %>%
  count(
    Genotype,
    exposed_group,
    name = "n_eyes"
  )

comp_rates <- first_complication_by_exposure %>%

  distinct(
    UID,
    complication_type,
    hydroxyurea_exposed_event
  ) %>%

  left_join(
    eye_df2 %>%
      select(
        UID,
        Genotype,
        exposed_group
      ) %>%
      distinct(),
    by = "UID"
  ) %>%

  count(
    Genotype,
    exposed_group,
    complication_type
  )


comp_rates <- comp_rates %>%

  left_join(
    denom,
    by = c(
      "Genotype",
      "exposed_group"
    )
```

```
    )
  ) %>%

  mutate(
    pct = 100 * n / n_eyes
  )



comp_table <- comp_rates %>%

  mutate(
    value = sprintf(
      "%d (%.1f%%)",
      n,
      pct
    )
  ) %>%

  select(
    Genotype,
    exposed_group,
    complication_type,
    value
  ) %>%

  tidyr::pivot_wider(
    names_from = complication_type,
    values_from = value
  )

comp_table%>%

  pivot_wider(
    names_from = exposed_group,
    values_from = -c(Genotype, exposed_group)
  ) %>%

  arrange(Genotype) %>%

  gt(
    rowname_col = "Genotype"
  ) %>%

  tab_header(
    title = md(
      "**Complication Type by Genotype and Hydroxyurea Exposure Status**"
    ),
    subtitle = md(
      "Values are presented as n (%) of eyes within each genotype-exposure group"
    )
  ) %>%

  tab_spanner(
    label = "Hydroxyurea Exposed",
    columns = ends_with("_exposed")
  ) %>%

  tab_spanner(
    label = "Hydroxyurea Unexposed",
    columns = ends_with("_unexposed")
  ) %>%



  cols_label(

    vitreous_hemorrhage_exposed = "VH",
    tractional_rd_exposed = "TRD",
    injection_exposed = "Injection",
    PRP_exposed = "PRP",
    vitrectomy_exposed = "vitrectomy",

    vitreous_hemorrhage_unexposed = "VH",
    tractional_rd_unexposed = "TRD",
    injection_unexposed = "Injection",
    PRP_unexposed = "PRP",
    vitrectomy_unexposed = "vitrectomy"
  ) %>%

  cols_align(
    align = "center",
    columns = everything()
  ) %>%

  tab_options(
    table.font.size = px(11),
    data_row.padding = px(4)
  )



### ANY

any_comp <- first_complication_by_exposure %>%

  distinct(UID) %>%

  mutate(any_complication = 1)




eye_df2 %>%

  distinct(
    UID,
    Genotype,
    exposed_group
  ) %>%
```

```
  )
  left_join(
    any_comp,
    by = "UID"
  ) %>%

  mutate(
    any_complication =
      ifelse(
        is.na(any_complication),
        0,
        1
      )
  ) %>%

  group_by(
    Genotype,
    exposed_group
  ) %>%

  summarise(

    eyes = n(),

    complications = sum(any_complication),

    pct = 100 * mean(any_complication),

    .groups = "drop"
  ) %>%


  mutate(
    complication_rate =
      sprintf(
        "%d (%.1f%%)",
        complications,
        pct
      )
  ) %>%

  select(
    Genotype,
    exposed_group,
    eyes,
    complication_rate
  ) %>%

  pivot_wider(
    names_from = exposed_group,
    values_from = c(
      eyes,
      complication_rate
    )
  )%>%

  gt(
    rowname_col = "Genotype"
  ) %>%

  cols_label(

    eyes_exposed = "Eyes",
    complication_rate_exposed = "Complications, n (%)",

    eyes_unexposed = "Eyes",
    complication_rate_unexposed = "Complications, n (%)"
  ) %>%

  tab_spanner(
    label = "Hydroxyurea Exposed",
    columns = c(
      eyes_exposed,
      complication_rate_exposed
    )
  ) %>%

  tab_spanner(
    label = "Hydroxyurea Unexposed",
    columns = c(
      eyes_unexposed,
      complication_rate_unexposed
    )
  ) %>%

  tab_header(
    title = md(
      "**Any Retinal Complication by Genotype and Hydroxyurea Exposure Status**"
    )
  ) %>%

  cols_align(
    align = "center",
    columns = everything()
  ) %>%

  tab_options(
    table.font.size = px(12),
    data_row.padding = px(4)
  )




```

Timing of diagnoses: Would it be possible to provide the number and percentage of cases where retinopathy was diagnosed before the first documented sickle cell disease diagnosis, stratified by genotype? This may help us assess whether coding/timing issues are affecting the genotype findings.

```{r}

testdf <- eye_df2 %>% select(UID,PatientDurableKey,Laterality,Date_retin_dx,Date_SC_dx) %>%
  mutate(days_retin_sc=as.numeric(Date_retin_dx-Date_SC_dx))


testdf %>% filter(days_retin_sc<0)



timing_tbl <- eye_df2 %>%
   mutate(days_retin_sc=as.numeric(Date_retin_dx-Date_SC_dx)) %>%

  distinct(
    UID,
    Genotype,
    days_retin_sc
  ) %>%

  mutate(
    retin_before_sc =
      ifelse(
        days_retin_sc < 0,
        "Yes",
        "No"
      )
  )



timing_tbl %>%

  group_by(Genotype) %>%

  summarise(

    Eyes = n(),

    RetinBeforeSC = sum(retin_before_sc == "Yes", na.rm = TRUE),

    Pct = 100 * RetinBeforeSC / Eyes,

    MedianDays = median(days_retin_sc, na.rm = TRUE),

    IQR_L = quantile(days_retin_sc, 0.25, na.rm = TRUE),

    IQR_U = quantile(days_retin_sc, 0.75, na.rm = TRUE),

    .groups = "drop"
  ) %>%

  mutate(

    `Retinopathy Before SCD, n (%)` =
      sprintf(
        "%d (%.1f%%)",
        RetinBeforeSC,
        Pct
      ),

    `Days Between SCD and Retinopathy Diagnosis` =
      sprintf(
        "%.1f (%.1f, %.1f)",
        MedianDays,
        IQR_L,
        IQR_U
      )
  )%>%

  select(
    Genotype,
    Eyes,
    `Retinopathy Before SCD, n (%)`,
    `Days Between SCD and Retinopathy Diagnosis`
  ) %>%

  gt(
    rowname_col = "Genotype"
  ) %>%

  tab_header(
    title = md(
      "**Timing of Sickle Cell Disease and Retinopathy Diagnoses by Genotype**"
    ),
    subtitle = md(
      "Negative values indicate retinopathy was documented before the first recorded sickle cell disease diagnosis"
    )
  ) %>%

  cols_label(
    Eyes = "Eyes",
    `Retinopathy Before SCD, n (%)` =
      "Retinopathy Before SCD, n (%)",
    `Days Between SCD and Retinopathy Diagnosis` =
      "Days Between Diagnoses, Median (IQR)"
  ) %>%

  cols_align(
    align = "center",
    columns = everything()
  ) %>%

  tab_source_note(
    source_note =
      "Retinopathy before SCD was defined as a retinopathy diagnosis date occurring prior to the first documented sickle cell disease diagnosis date."
  )
```

```
timing_tbl %>%
  mutate(
    timing_group = case_when(
      days_retin_sc < 0 ~ "Retinopathy before SCD",
      days_retin_sc <= 365 ~ "Within 1 year",
      TRUE ~ ">1 year after SCD"
    )
  ) %>%
  count(Genotype, timing_group) %>%
  group_by(Genotype) %>%
  mutate(
    pct = 100*n/sum(n)
  )


timing_tbl %>%
  mutate(
    timing_group = case_when(
      days_retin_sc < 0 ~ "Retinopathy before SCD",
      days_retin_sc <= 365 ~ "Within 1 year",
      TRUE ~ ">1 year after SCD"
    )
  ) %>%
  count(Genotype, timing_group) %>%
  group_by(Genotype) %>%
  mutate(
     N = sum(n),
    pct = 100 * n / sum(n),
    stat = sprintf("%d (%.1f%%)", n, pct)
  ) %>%
  ungroup() %>%
  select(Genotype, timing_group, stat,N) %>%
  pivot_wider(
    names_from = timing_group,
    values_from = stat,
    values_fill = "0 (0.0%)"
  ) %>%
  gt(rowname_col = "Genotype") %>%
  tab_header(
    title = "Timing of Retinopathy Relative to SCD Diagnosis"
  ) %>%
  cols_label(
    `Retinopathy before SCD` = "Retinopathy Before SCD",
    `Within 1 year` = "Within 1 Year of SCD",
    `>1 year after SCD` = ">1 Year After SCD"
  )
```